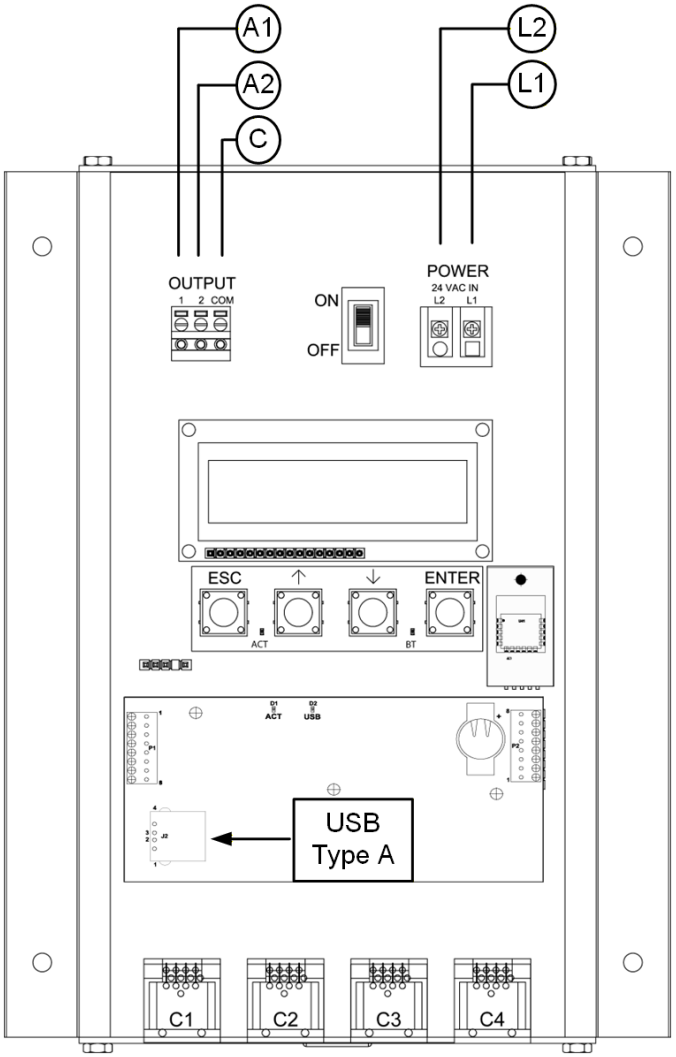
# Advantage IV (A4) HTU116-P STARTUP GUIDE




**HTU116-P Parameters - Factory Defaults and Optional Settings/Ranges**

| Description | Parameter | Default | Optional Settings/Ranges | Units |
|---|---|---|---|---|
| System of Units | SYS | I-P (US customary) | SI (metric) | |
| Airflow Calculation Method | AIRFLOW | ACT (actual) | STD (standard mass flow) | |
| Altitude (for actual flow correction) | ALT | 0 | AUTO, 0 to 20000 [0 to 6000] | ft [m] |
| Low Limit Airflow Cutoff | LLIMIT | 0 FPM | 0 to 500 FPM [0.0 to 2.5 m/s] | |
| Area | AREA | *{Order Area}* | 0.00 to 9999.99 [0.000 to 999.999] | sq ft [sq m] |
| AO1 and AO2 Type | AOUT | 4-20mA | 0-10V, 0-5 V | |
| AO1 Assignment | AO1 ASGN | AF (Airflow) | None | |
| AO1 Unit of Measure | AO1 UM | FPM [m/s] | CFM [L/s] | |
| AO1 Minimum Scale Reading | AO1 MS | 0 | None | FPM [m/s] |
| AO1 Full Scale Reading | AO1 FS | 5000 [25.0] | 100 to 15000 [0.5 to 75.0] | FPM [m/s] |
| AO2 Assignment | AO2 ASGN | TEMP (Temperature) | ALRM (Alarm) or TRBL (System Trouble) | |
| AO2 Unit of Measure | AO2 UM | F [C] | None | °F [°C] |
| AO2 Minimum Scale Reading | AO2 MS | -20 [-30] | -50 to 160 [-50 to 70] | °F [°C] |
| AO2 Full Scale Reading | AO2 FS | 160 [70] | -50 to 160 [-50 to 70] | °F [°C] |
| Datalogger | Requires a Type A USB Datalogger "thumb drive" (see below) | | | |
| USB Port (turn off before removal) | USB WRITE | ON | OFF | |

Refer to the O&M Manual for more information and/or additional parameter defaults, settings and ranges.

STARTUP INSTRUCTIONS:

1. Verify that the sensor probes are located where they meet EBTRON published installation guides.
2. Verify that the probes are properly spaced with the airflow arrows pointing in the direction of airflow.

⚠ *Improperly installed probes will compromise the installed accuracy of the device and degrade system performance.*

3. Verify that the transmitter is installed and wired in accordance with the *HTU116-P Wiring Guide* provided with the transmitter and power is provided to the transmitter.
4. Make sure the ductwork is clean and free of debris prior to fan startup.
5. Move the power switch to the "ON" position. Power-up faults, if detected, are displayed on the LCD. If any power up faults are detected, resolve all conflicts or contact EBTRON customer service at 1-800-232-8766 before proceeding.

⚠ *If extension cables have been added, the extension cable length must be entered into the transmitter. Refer to the Operations and Maintenance Manual for more information.*

6. The transmitter is fully functional as a factory calibrated airflow and temperature measurement device in I-P units (ft, FPM, CFM °F). Airflow (CFM) and temperature (°F) are displayed on the LCD.

ⓘ *If SI units are required, refer to the Operations and Maintenance Manual.*

ⓘ *The factory default airflow output is set to actual airflow (FPM, CFM). If standard (mass) airflow (SFPM, SCFM) is required, refer to the Operations and Maintenance Manual.*

7. Verify that the location (if provided) on the hang-tag matches the actual location where sensor probes are installed to optimize implementation of the EB-Link Reader. The NAME parameter stored in transmitter will match the location on the hang-tag or will default to transmitter serial number if no location was provided or location does not match on all sensor probes. If the location does not match or is blank on hang-tag and requires entry, modify the NAME parameter in the transmitter.

ⓘ *If the NAME parameter must be changed, refer to the Operations and Maintenance Manual.*

ⓘ *The NAME parameter will be displayed on the EB-Link Reader.*

8. Verify that the area on the hang-tag matches the actual area of the duct or opening where the probes are located (less any internal insulation). If the area is different, modify the area parameter stored in the transmitter and use the correct area for any external conversion calculations from FPM to CFM.

⚠ *Failure to use the correct area <u>will</u> result in volumetric airflow (CFM) measurement error and degrade system performance. If the area parameter must be changed, refer to the Operations and Maintenance Manual.*

9. If analog output signals are used continue to step 10, otherwise skip to step 14.
10. The output signal type and range (4-20 mA, 0-5 VDC or 0-10VDC) of AO1 and AO2 is determined by the AOUT parameter. The transmitter is factory set to 4-20mA.

⚠ *The 4-20mA is "4-wire type" and not loop powered. Do not apply any excitation voltage to the output of the transmitter.*

11. Verify that the transmitter is configured to match the analog input requirements of the host controller. Press the ESC and ↑ buttons simultaneously to display the transmitter setting for the AOUT parameter. If the AOUT parameter is not correct, press the ENT button and use the ↑ and ↓ buttons to set AOUT. Press the ENT button to execute and display the change. Press the ESC button to return to normal operation.
12. The analog output signal for airflow (AO1) is linear. The minimum scale reading (0% output) of the airflow signal is fixed at 0 and the full scale reading (100% output) is factory set to 5,000 FPM.

*Multiply the default full scale velocity (FPM )by the correct area of the measurement location to determine the full-scale or span (CFM ) for the B.A.S. to avoid field configuration. EBTRON airflow measurement device accuracy is percent-of-reading. Changing the full scale reading does not affect measurement accuracy.*

ⓘ *If custom airflow scaling or unit of measure are required, refer to the Operations and Maintenance Manual.*

13. The analog output signal for temperature (AO2) is linear. The minimum scale reading (0% output) is set to -20 ºF and full scale reading (100% output) is set to 160 ºF.

ⓘ *If custom temperature scaling is required, refer to the Operations and Maintenance Manual.*

ⓘ *AO2 can be configured for a high/low airflow alarm or system status alarm. Refer to the Operations and Maintenance Manual for more information.*

14. If USB memory device connection is required continue to step 15, otherwise skip to step 23.
15. Transmitters with a USB data logger log sensor node data at 5-minute intervals using Universal Time Coordinated (UTC) based on an onboard real-time clock, whenever power is applied to the transmitter. Data files are automatically appended on power-up.

    The comma separated values (.CSV) filenames and file details are as follows:

    - DATAF.CSV - Average airflow and individual sensor node airflow
    - DATAT.CSV - Average temperature and individual sensor node temperature

ⓘ *For standard transmitters equipped with EB-Link Bluetooth® low energy interface, the time zone and interval can be modified by the user using EB-Link software.*

16. To insert a USB memory device ("thumb drive") for data logging continue to step 17, otherwise skip to step 20.
17. Install the USB memory device into the USB connector on the option card to start logging data.

*It is a good practice to set the transmitter power switch to the "OFF" position before inserting the USB memory device.*

18. The USB port must be enabled to log data.
19. Enable the USB WRITE parameter to start logging data. Press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button. The USB WRITE parameter is visible. Press the ENT button and use the ↑↓ buttons to select ON. Press the ENT button to save the selection. Press the ESC button twice to return to normal operation.

ⓘ *Transmitters are shipped with the USB port parameter USB WRITE set to "ON".*

20. To remove the USB memory device and stop data logging continue to step 21, otherwise skip to step 23.
21. Disable the USB WRITE parameter stop logging data. Press the ↑↓ arrow buttons simultaneously to enter the MAIN MENU. The SETTINGS menu is displayed. Press the ENT button to select the top of the SETTINGS submenu category. Press the ↓ button until the USB submenu category is visible. Press the ENT button. The USB WRITE parameter is visible. Press the ENT button and use the ↑↓ buttons to select OFF. Press the ENT button to save the selection. Press the ESC button twice to return to normal operation.

⚠ Always set the USB WRITE parameter to "OFF" before removing the USB memory device to avoid data loss/and or damage.

22. Remove the USB memory device from the USB port, if desired.

*It is a good practice to set the transmitter power switch to the "OFF" position before removing the USB memory device.*

23. Startup is complete! If additional customization is desired, consult the *Operation and Maintenance Manual*.

## VERIFICATION

Many installations require third-party airflow verification. If the airflow measuring device is within the measurement uncertainty of the verification technique, EBTRON strongly recommends that no field adjustment correction is made. EBTRON airflow measurement devices are factory calibrated to NIST traceable standards. Field adjustment is not recommended when installed in accordance to published guidelines.

If field adjustment is required, refer to the *Operation and Maintenance Manual*.

ⓘ *If minimum placement guidelines cannot be achieved, installed accuracy may be compromised. Transmitters can be field adjusted to match a third-party measurement. Adjusted field measurements typically result in comparative readings within ±3% of the third-party measurement. Be advised that the third-party measurement may have uncertainties greater than or equal to ±10% and should only be used to adjust the airflow measurement device if the probes do not meet minimum placement requirements or the discrepancy is greater than the uncertainty of the third-party source.*

## NEED MORE HELP?

<u>EBTRON Customer Service</u>

For toll-free factory support call 1-800-2EBTRON (1-800-232-8766), Monday through Thursday 8:00 AM to 4:30 PM and Friday 8:00 AM to 2:00 PM eastern time.

<u>Your Local EBTRON Representative</u>

Visit EBTRON.com for the name and contact information of your local representative.